MSBGen

Generative Image Reconstruction from Most-Significant Bitplanes

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Motivation: Can We Generate What We Don't Transmit?

- Conventional 8-bit images require 8 bits/pixel per channel.
- The Most Significant Bit (MSB) contains coarse intensity and structural information.
- Can we transmit only the MSB and generate the remaining information at the receiver?

$$I = 128B_7 + 64B_6 + \cdots + B_0$$

$$\underbrace{I_{\text{top}} = 128B_7}_{\text{Transmitted}} \qquad \underbrace{R = I - 128B_7}_{\text{Missing information}}$$

$$\boxed{\hat{I} = 128B_7 + \hat{R}}$$

Core Research Question

Can conditional generative models reconstruct a perceptually useful full image from an extremely limited representation?

Key challenge: Multiple original images can share the same MSB plane. This is an inherently one-to-many reconstruction problem.

Existing Research

- **Punnappurath & Brown (2020) – Bitplane-wise Bit-depth Recovery** [1]
  - Uses a CNN to predict missing lower bitplanes sequentially.
  - Focuses on recovering bit depth from partially available bitplanes.
- **Chang et al. (2021) – Progressive Medical Image Transmission** [2]
  - Uses a bank of GANs for progressive medical image transmission.
  - Generates image refinements at successive transmission stages.
- **Han et al. (2025) – Bit Plane Decomposition for INR** [3]
  - Uses bitplane decomposition within an implicit neural representation.
  - Aims at lossless image representation.

Research Gap

- Existing bitplane-based methods primarily focus on:
  - Sequential recovery of missing lower bitplanes.
  - Lossless representation using bitplane decomposition.
  - Progressive image transmission through multiple stages.

- **What is less explored?**
  - Direct reconstruction of the complete image using only the top k most significant bitplanes.
  - Learning the missing image information as a residual rather than explicitly predicting each lower bitplane.
  - Comparing regression-based and generative approaches for this task.

- **My objective:** Given the top k MSB planes, reconstruct a visually plausible full 8-bit grayscale or RGB image.

From Regression to Generative Models

1. **Regression-based reconstruction**
   - CNN, U-Net and residual U-Net models directly predict the missing image residual from the available MSB planes.
   - They learn a deterministic mapping from the input bitplanes to the reconstructed image.
   - These models provide strong pixel-level reconstruction baselines.

2. **Why explore generative models?**
   - Multiple plausible images may correspond to the same limited MSB information.
   - A deterministic prediction may not capture the uncertainty in the missing details.
   - Generative models allow the reconstruction of diverse, plausible high-frequency details.

From Regression to Generative Models

Generative approaches explored

- **Conditional Diffusion:** Iteratively denoises a residual conditioned on the available MSB planes.
- **Flow Matching:** Learns a continuous transformation from noise to a plausible residual.
- **Conditional Normalizing Flow:** Learns an invertible mapping to model the conditional residual distribution.

Hybrid approach

- Residual U-Net provides an initial reconstruction.
- Flow matching models a residual correction.
- Diffusion further refines the generated result.

Goal: Combine accurate reconstruction with perceptually plausible details.

Results: RGB Kodak Benchmark

Experimental setup

- Training: 800 RGB DIV2K; Evaluation: 24 RGB Kodak images.
- k = number of retained MSB planes per channel.
- Compare sequential bitplane CNN (D4) against the proposed three-stage hybrid.

Quantitative comparison

	PSNR (dB) ↑		**LPIPS** ↓		**KID** ($\times 10^3$) ↓	
k	D4	Hybrid	D4	Hybrid	D4	Hybrid
1	18.40	**18.56**	0.4481	**0.4223**	87.81	**53.35**
2	**27.00**	26.48	0.2367	**0.1898**	50.51	**18.13**
3	**33.89**	32.52	0.0964	**0.0772**	21.43	**6.12**

- Sequential bitplane CNN achieves higher PSNR at $k = 2, 3$.
- Hybrid consistently improves LPIPS and KID across all three bit-depth settings.

Original Image



Full 8-bit image

First MSB Plane



Binary representation of the first MSB

MSB-only Image



First MSB retained; all other planes set to zero

U-Net



PSNR (dB) ↑	18.01
SSIM ↑	0.65
LPIPS ↓	0.3571
DISTS ↓	0.2273
MAE ↓	26.80
Time (s) ↓	0.126

Flow Matching



PSNR (dB) ↑	18.07
SSIM ↑	0.6323
LPIPS ↓	0.3461
DISTS ↓	0.1851
MAE ↓	26.03
Time (s) ↓	0.606

Diffusion



PSNR (dB) ↑	17.87
SSIM ↑	0.6287
LPIPS ↓	0.3403
DISTS ↓	0.1924
MAE ↓	26.45
Time (s) ↓	0.605

U-Net Hybrid



PSNR (dB) ↑	18.04
SSIM ↑	0.6423
LPIPS ↓	0.3467
DISTS ↓	0.1844
MAE ↓	26.11
Time (s) ↓	1.086

ViT Hybrid



PSNR (dB) ↑	17.33
SSIM ↑	0.6268
LPIPS ↓	0.3557
DISTS ↓	0.2030
MAE ↓	27.28
Time (s) ↓	1.365

Bitdepth D4



PSNR (dB) ↑	18.03
SSIM ↑	0.6574
LPIPS ↓	0.3445
DISTS ↓	0.1781
MAE ↓	26.26
Time (s) ↓	2.001

Conclusion and Takeaways

1. **Feasibility:** Useful full-image reconstruction is possible even when only a small number of MSB planes are retained.
2. **Hybrid modeling:** Combining deterministic regression with flow matching and diffusion improves perceptual metrics over standalone bitplane CNN recovery.
3. **Perception vs. distortion:** No single method wins across all metrics. Better perceptual similarity can come with a small PSNR trade-off.
4. **Potential application:** Generative reconstruction offers a direction towards extremely low-bitrate image communication.

Current Limitations

Evaluation is based on a single training run and 24 Kodak test images. Multiple seeds, additional datasets, actual entropy-coded bitrate measurements and inference scaling remain important next steps.

References

1. Punnappurath, A. and Brown, M. S. (2020). *A Little Bit More: Bitplane-Wise Bit-Depth Recovery.* IEEE TPAMI. arXiv:2005.01091

2. Chang et al. (2021). *Progressive Transmission of Medical Images via a Bank of GANs.* DOI: 10.1155/2021/9917545

3. Han et al. (2025). *Towards Lossless Implicit Neural Representation via Bit Plane Decomposition.* arXiv:2502.21001

My code is available at: `https://github.com/DevadattaP/MSBGen`